Audio-Augmented Prompting for Humor Explanation in Open-Source Language Models

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***Abstract*—Humor understanding remains a challenging task for large language models, particularly when interpretation depends on ambiguity and spoken realization. Although such models perform strongly on many natural language understanding tasks, humor often relies on cues that are not fully recoverable from text alone, including phonetic similarity and pronunciation. This paper investigates whether incorporating audio can improve humor explanation in locally deployable multimodal systems. We compare text-only, audio-only, and text-plus-audio prompting settings and evaluate the generated explanations using a comparison-based judging framework, alongside a complementary label-based evaluation. The results indicate that audio is most useful when combined with text and when the humorous effect is closely tied to pronunciation-based ambiguity, while its benefits are less consistent for humor that does not primarily depend on phonetic cues. Overall, the findings suggest that audio can serve as a useful complementary signal for humor understanding, but not a universally beneficial one.**

***Index Terms*—humor understanding, multimodal prompting, text-to-speech, prosody, explanation generation**

I. Introduction

Recent advances in large language models have enabled strong performance across many natural language understanding tasks [1]. However, humor understanding remains challenging, especially when linguistic ambiguity and phonetic cues are central. Similar observations have been made in related multimodal affective tasks such as sarcasm recognition, where combining textual and acoustic cues improves performance [2]. More broadly, linguistic ambiguity is a well-known source of humor, including lexical and syntactic ambiguity in short texts such as newspaper headlines [3]. In pun-based humor, pronunciation and prosody can provide information not always recoverable from written text alone.

The rest of the paper is organized as follows. Section II reviews related work on humor understanding, puns, and multimodal language processing. Section III describes the dataset and task formulation. Section IV presents the experimental pipeline, including text-only and text-plus-audio explanation generation, JSON normalization, and judge-based evaluation. Section V outlines the models and implementation details. Section VI reports the results and analysis. Section VII concludes and discusses future work.

II. Related Work

Computational humor research has long emphasized ambiguity as a central mechanism underlying jokes and puns. Early linguistic theories characterize humor as arising from the coexistence of multiple semantic interpretations within a single utterance [4]. Building on this perspective, formal semantic and pragmatic analyses of puns highlight the role of phonological similarity and lexical ambiguity in enabling dual readings [5]. In computational settings, humor detection and interpretation have been explored using supervised and neural approaches, including benchmarks such as SemEval-2017 Task 7, which operationalizes pun detection and sense interpretation in English text [6].

More recently, large language models have demonstrated strong capabilities in general natural language understanding, yet their performance on humor remains inconsistent, particularly when jokes rely on subtle ambiguity or world knowledge [7]. Parallel developments in multimodal language understanding suggest that integrating acoustic information can improve performance in tasks such as emotion recognition and spoken language understanding, where prosody and delivery provide complementary cues to text [8]. These findings motivate examining whether similar multimodal benefits extend to pronunciation-dependent humor, especially under reproducible, locally deployable model settings. This study builds on these lines of research by examining multimodal humor understanding in the specific context of pun interpretation. More precisely, it extends prior work by investigating whether the inclusion of speech-related information can improve the interpretation of ambiguity-driven humor in large language models.

III. Methods

A controlled evaluation is conducted to determine whether access to audio information improves humor understanding and explanation in locally deployed open-source language models. Given the limited capacity of such models relative to large proprietary systems, a simple multimodal prompting framework is adopted. Few-shot demonstrations and explicit chain-of-thought prompting are not used. Instead, all exper-

iments are performed in a zero-shot setting with controlled output formatting.

As illustrated in Fig. 1, the pipeline is organized into four stages: (1) explanation generation from text-only input, (2) explanation generation from combined text and audio input, (3) post-processing and normalization of model outputs into a strict JSON representation, and (4) comparative evaluation using a separate judging model.

A. Audio generation

Spoken realizations of all jokes are generated to expose pronunciation-related cues relevant to humor understanding. An off-the-shelf text-to-speech system is employed. Specifically, Piper TTS [9] is used as a lightweight open-source engine suitable for local deployment, and audio is synthesized directly from the original joke text. No additional supervision is provided, including emphasis, timing, or prosody annotations. The synthesized audio is used only as an auxiliary input intended to reveal pronunciation-based ambiguity.

B. Prompting

All explanation prompts are formulated in a zero-shot configuration. Each prompt includes a concise definition of puns and non-puns, together with instructions to determine whether the input constitutes a pun and to describe the linguistic mechanism when applicable. Few-shot demonstrations and explicit chain-of-thought prompting are omitted because preliminary trials indicate that smaller local models do not produce stable and consistent outputs under these settings.

Two explanation conditions are evaluated in the main setup: (1) **Text-Only** - the model receives only the written joke, and (2) **Text-Plus-Audio** - the model receives the written joke and its synthesized spoken form within a single prompt. For the extended evaluation on SemEval, an additional **Audio-Only** condition is also considered, in which the model receives only the synthesized spoken form of the joke. To ensure compatibility with text-only ground truth annotations, the prompt instructs the model to avoid referencing the presence or absence of audio in the generated explanation.

C. Output normalization and evaluation

All generated explanations are converted to a strict JavaScript Object Notation (JSON) format to enforce consistency and enable automated comparison. Outputs that do not conform to the required format are marked as invalid.

To assess explanation quality, a separate judging model is employed. For pairwise comparisons, the judge receives the original joke text and the two explanations produced under the text-only and text-plus-audio conditions and determines which explanation more accurately identifies whether the input constitutes a pun and more effectively explains the linguistic mechanism. The judging instructions discourage positional bias and verbosity bias and emphasize correctness, clarity, and accurate identification of wordplay. A tie decision is permitted when both explanations are comparable. No audio input is provided to the judging model.

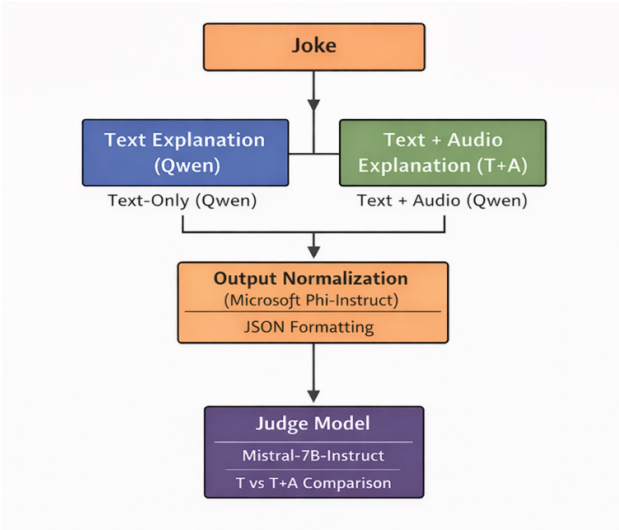

Fig. 1. Pipeline overview: text-only and text-plus-audio explanation generation, JSON normalization, and judge evaluation.

IV. Datasets and Models

The proposed approach is evaluated on three humor datasets that differ in dataset size, annotation quality, and the availability of human-written explanations.

1) SemEval 2017 Task 7: The dataset [6] comprises 810 homographic puns, 647 heterographic puns, and 1,077 non-puns, with pun instances explicitly partitioned by type. This structure makes the dataset well-suited for evaluating explanation quality and for analyzing the linguistic mechanisms underlying pun-based humor.

2) Context-Situated Puns: The Context-Situated Puns dataset contains 821 homographic puns and 1,739 heterographic puns with human annotations specifying the pun word and its intended interpretations. Each instance consists of a single sentence containing a pun, supporting evaluation of whether the model can infer and explain the underlying linguistic ambiguity from sentence-level context [10].

3) ExplainTheJoke: To evaluate explanation quality beyond pun-based humor, the ExplainTheJoke dataset is used in its entirety. The dataset contains jokes spanning multiple humor categories and does not include structured annotations. For each joke, the judge evaluates the text-only explanation and the text-plus-audio explanation *independently* on a five-point scale based on correctness, clarity, and accurate identification of the humor mechanism, while penalizing hallucinated or irrelevant reasoning. No audio input is provided to the judging model. Pairwise preferences are then derived by comparing the two scores; equal scores are counted as ties.

A. Models

All experiments are conducted using freely available locally deployable language models.

Explanation generation: Both text-only and text-plus-audio explanations are generated using Qwen2-Audio [11], which supports audio-conditioned generation while remaining

deployable in a local environment. The same model and prompt structure are used for both conditions to ensure a controlled comparison, with the only difference being the presence or absence of audio input. Additionally, Qwen3-Omni-30B-A3B-Instruct [12] is employed in the SemEval-based experiments, where explanations are generated under text-only, audio-only, and text-plus-audio conditions.

Output normalization: Generated explanations are normalized into a strict JSON format using Microsoft Phi Instruct [13]. Outputs that fail to conform to the required format are marked as invalid and excluded from further analysis.

Evaluation judge model: Explanation quality is assessed using Mistral 7B Instruct v0.3 [14], employed as an independent judging model. Using a separate model for evaluation reduces the risk of a model judging its own outputs.

V. Results

A. SemEval 2017 Task 7

Evaluation on SemEval is performed using a comparison-based judging framework. In the initial experiment, a balanced subset of 250 homographic and 250 heterographic puns is sampled, and for each instance two explanations are generated under text-only and text-plus-audio prompting conditions. The resulting pairwise preference outcomes for this subset-based setting are reported in Table I. The evaluation is then extended to the full dataset using Qwen3-Omni-30B-A3B-Instruct. In this setting, two pairwise comparisons are conducted for each instance, namely text-only versus audio-only and text-only versus text-plus-audio, and the corresponding preference outcomes are reported in Table II. In addition to the comparison-based analysis, the generated outputs are also evaluated as classification decisions against the gold labels of the original SemEval-2017 Task 7 task formulation, with the resulting classification metrics summarized in Table III.

Example (Audio improves explanation): *Joke:* A chicken farmer's favorite car is a coupe.

Text-only: The text uses the word "coupe" which sounds like "couple," creating a double meaning between a car type and a pair of chickens.

Text+audio: The text uses the word "coupe" which sounds like "coop," creating a double meaning that plays on the context of a chicken farmer.

Conclusion: The text-only explanation misinterprets the intended wordplay, while the audio condition enables the correct identification of the phonetic relationship, leading to an accurate explanation of the pun.

Example (No improvement or degradation): *Joke:* I had some back trouble, but it's all behind me now.

Text-only: The phrase uses a play on words with "behind me," which has a double meaning—referring both to physical location and metaphorical distance—creating a pun with "back trouble."

Text+audio: The phrase "back trouble" and "behind me" create a pun based on a double meaning of the word "back." The first meaning refers to physical discomfort in the back, while the second meaning uses "behind me" to indicate that the problem is no longer present. The wordplay hinges on the homophonic relationship between "back" and "behind."

Conclusion: The text-only explanation provides a clear and accurate account of the pun, while the audio condition introduces inaccurate and less relevant reasoning without improving the explanation.

TABLE I
SemEval results under pairwise judging (valid outputs only).

Split (Valid N)	Text wins	Audio wins	Tie
Heterographic (188)	41.5%	58.0%	0.5%
Homographic (194)	47.4%	49.5%	3.1%

TABLE II
SemEval Pairwise Results (Qwen3-Omni)

Comparison	Type	Text	Other	Same
Text vs. Audio	Heterographic	71.38%	27.21%	1.41%
Text vs. Audio	Homographic	73.22%	21.69%	4.98%
Text vs. T+A	Heterographic	46.21%	48.96%	4.84%
Text vs. T+A	Homographic	52.68%	36.82%	10.91%

B. Context-Situated Puns

For Context-Situated Puns, evaluation is conducted using the same judging framework, but explanations are scored independently rather than compared directly. A subset of 300 examples is sampled, covering both homographic and heterographic puns. For each example, two explanations are generated under the text-only and text-plus-audio conditions. The judging model assigns each explanation a score on a five-point scale (1–5) based on correctness, clarity, and accurate identification of the linguistic mechanism. Pairwise preferences are then derived by comparing the two scores, with equal scores counted as ties. A paired t-test on the per-example judge scores indicates that the difference between text-only and text-plus-audio explanations is not statistically significant ($t(298) = 1.62$, $p = 0.106$). This suggests that the observed difference in mean scores is likely due to chance, which is consistent with the high proportion of ties.

C. ExplainTheJoke

For ExplainTheJoke, the judge scores each explanation independently on a five-point scale. A pairwise preference is then derived by selecting the explanation with the higher score; equal scores are counted as ties. Table V reports the resulting preference distribution. A paired t-test on the per-example judge scores shows that the difference between text-only and text-plus-audio explanations is not statistically significant ($t(376) = 0.94$, $p = 0.348$). This result aligns with the high proportion of ties, indicating minimal difference between the two conditions.

Overall, the results indicate that the effect of audio is dependent on the type of humor and the manner in which it is incorporated. Audio appears to be most beneficial when combined with text and when the humorous effect is closely tied to pronunciation-based ambiguity, as seen most clearly in

TABLE III
CLASSIFICATION RESULTS ON THE FULL SEMEVAL-2017 TASK 7 DATASET FOR THE DIFFERENT INPUT CONDITIONS.

Setting	Pun Type	Samples	Correct	Accuracy	Precision	Recall	F1 Score
Text + Audio	Heterographic	1738	1267	0.7290	0.7330	0.9750	0.8369
Text + Audio	Homographic	2198	1605	**0.7302**	0.7303	0.9847	**0.8386**
Audio Only	Heterographic	1728	1225	0.7089	0.7139	0.9887	0.8291
Audio Only	Homographic	2205	1576	0.7147	0.7143	**0.9981**	0.8327
Text Only	Heterographic	1729	1217	0.7039	0.7284	0.9281	0.8162
Text Only	Homographic	2228	**1618**	0.7264	**0.7382**	0.9491	0.8305

TABLE IV
CONTEXT PUNS: PAIRWISE PREFERENCES

Outcome (Valid N=299)	Count	Percent
Tie (scores equal)	232	77.6%
Text > Audio	53	17.7%
Audio > Text	14	4.7%
Score statistics (1–5)	**Mean**	**Median**
Text-only	3.696	3.0
Text + Audio	3.428	3.0

TABLE V
EXPLAINTHEJOKE: PAIRWISE PREFERENCES

Derived preference	Count	Percent
Text-only (score higher)	63	16.7%
Text + Audio (score higher)	39	10.3%
Tie (scores equal)	275	72.9%

the heterographic pun setting. At the same time, the weaker performance of the audio-only condition and the large number of ties in the broader humor datasets suggest that speech does not uniformly improve explanation quality. These findings support the view that auditory information functions primarily as a complementary cue, whose usefulness depends on whether the underlying humor mechanism is phonologically grounded.

VI. DISCUSSION AND CONCLUSION

Across the considered datasets, the effect of incorporating audio is mixed and appears to depend on both the type of humor and the reliability of the local generation pipeline. The clearest benefits emerge in pronunciation-dependent cases, particularly heterographic puns, where the spoken form can reveal phonetic relationships that are less explicit in text alone. In such cases, audio functions as a useful complementary signal that can help the model recover the ambiguity underlying the humorous effect and produce more informative explanations.

At the same time, the results show that this benefit is not consistent across all humor types. For examples that depend less on phonetic ambiguity, the addition of audio yields limited or inconsistent gains, and text-only prompting often remains equally effective. A likely explanation is that many humorous instances rely more on semantic contrast, contextual inference, or background knowledge than on acoustic realization. In these settings, synthesized speech may provide little additional information and can also introduce variability through imperfect pronunciation, prosody, or timing. The weaker performance of audio-only prompting further supports this interpretation, suggesting that speech alone is generally insufficient for reliable humor explanation when important lexical and orthographic cues are absent.

Taken together, these findings suggest that multimodal prompting with locally deployable models is promising, but not universally beneficial. Audio appears to be most useful when it complements text and when the humorous effect is grounded in sound-based ambiguity. Future work should therefore explore higher-quality speech synthesis, more controlled prosody, and prompting strategies that make better use of auditory information while preserving stable and consistent outputs across different humor types.

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